Movie Memory Experiment

Event Segmentation and Boundary Effects on Recognition Memory

Vibe Coders

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GitHub Repository: https://github.com/PranjalSK03/BRSM_Project

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1 Introduction

This study investigates how the temporal position of a video clip's ending relative to natural event boundaries affects recognition memory for individual frames. Understanding how humans segment continuous experiences into discrete events has profound implications for memory, attention, and perception. Our experiment tests whether disrupting natural event boundaries impairs memory consolidation by comparing two experimental conditions: videos that end at natural boundaries versus videos that are artificially truncated before reaching those boundaries.

1.1 Theoretical Background

1.1.1 Event Segmentation Theory

Event Segmentation Theory (EST) (Zacks et al., 2007) proposes that humans spontaneously segment continuous experience into discrete events. The cognitive system maintains predictive models of what will happen next. When predictions fail (e.g., scene changes, action shifts), the system experiences **prediction error**, signaling **event boundaries**. These boundaries trigger enhanced attention and memory encoding.

1.1.2 The Boundary Advantage Effect

The **Boundary Advantage Effect** (Swallow et al., 2009) demonstrates superior memory for information at event boundaries compared to event middles. At boundaries, the cognitive system "saves" the current event representation to long-term memory while preparing to encode a new event, enhancing memory traces.

1.1.3 Predictions for Video Memory

If natural event boundaries enhance memory encoding, then disrupting these boundaries should impair recognition memory. Our experiment manipulates whether participants see complete events (ending at natural boundaries) or truncated events (ending abruptly before boundaries). We predict that truncation will prevent the normal boundary-related memory enhancement, leading to worse recognition for frames from truncated videos. Moreover, the effect should be especially strong for frames that would have appeared near the (now absent) boundary.

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2 Experimental Design and Hypotheses

2.1 Design Overview

Participants were randomly assigned to one of two between-subjects conditions: - **NB (Natural Boundary) Condition**: Videos ended at natural event boundaries (e.g., a person completing an action, a scene concluding) - **AB (Abrupt Boundary) Condition**: Videos were truncated 1-5 seconds before the natural boundary, creating artificial endings mid-event

All participants viewed 40 short video clips (17-57 seconds each) depicting everyday activities. After the viewing phase, participants completed a forced-choice recognition task for 40 trials. Each trial presented two video frames: one frame actually shown in the video (target) and one similar frame not shown (lure). Participants selected which frame they had seen and rated their confidence (1-5 scale).

2.2 Frame Types

Each recognition trial tested one of two frame types: - **BB (Before Boundary)**: Frames from 0-3 seconds before the natural event boundary - **EM (Event Middle)**: Frames from the middle of the event, far from boundaries

This within-subjects manipulation allowed us to test whether boundary effects are specific to boundary-adjacent frames or generalize across the entire event.

2.3 Study Hypotheses

Hypothesis 1 (Main Effect): NB participants will demonstrate higher overall recognition accuracy than AB participants. Rationale: Natural boundaries allow complete event model updating and memory consolidation, while abrupt truncation interrupts this process.

Hypothesis 2 (Boundary Frame Advantage): The NB advantage will be significantly stronger for BB frames than for EM frames. Rationale: BB frames are most proximal to the disrupted boundary and should be most affected by boundary presence/absence.

Hypothesis 3 (Null Effect on EM Frames): NB and AB conditions will not differ significantly on EM frame accuracy. Rationale: Frames from event middles are temporally distant from the disrupted boundary and should be minimally affected by the manipulation.

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3 Data and Data Cleaning

3.1 Data Structure and Cleaning Pipeline

Dataset: **170 participants** $\times$ 40 trials = 6800 observations. Key variables: participant_id, Condition (NB/AB), FrameType (BB/EM), Accuracy (0/1), RT (seconds), Confidence (1-5).

Cleaning steps: (1) Excluded N=1 participants with <20 trials; (2) Converted variables to numeric; (3) Classified frames as BB/EM from filenames; (4) Removed RT outliers (cutoff = Mean + 3SD = 17.15s); (5) Listwise deletion on critical variables; (6) Aggregated to participant level (N=170) for statistical independence.

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4 Variables and Descriptive Analysis

4.1 Variable Types

IVs: Condition (between-subjects, nominal: NB/AB); FrameType (within-subjects, nominal: BB/EM). **DVs:** Accuracy (ratio: 0-1), RT (ratio: seconds), Confidence (ordinal: 1-5). **Demographics:** Age (ratio: years), Gender (nominal: Male/Female).

Dependent Variables (Outcomes): - **Accuracy**: Continuous ratio variable (proportion correct, range 0-1 after aggregation). Primary outcome. - **Response Time (RT)**: Continuous ratio variable (seconds). Secondary outcome, measured from frame presentation to response keypress. - **Confidence**: Ordinal variable (1-5 scale, higher = more confident). Self-reported certainty.

Demographic Variables (Participant Characteristics): - **Age**: Continuous ratio variable (years) - **Gender**: Nominal categorical variable {Female, Male, Other} - **Handedness**: Nominal categorical variable {Right, Left, Ambidextrous} - **Vision**: Nominal categorical variable {Normal, Corrected-to-normal, Other}

4.2 Sample Characteristics

Final Sample Size: N = 170 participants (excluded 1 with incomplete data)

Condition Assignment: - NB Condition: N = 89 participants - AB Condition: N = 81 participants

The randomization produced reasonably balanced groups for subsequent independent-samples comparisons.

4.3 Measures and Data Processing

Trials with RT > Mean + 3SD were excluded as outliers. Data were aggregated to the participant level to satisfy statistical independence assumptions.

4.4 Demographic Summary Statistics

4.5 Sample Characteristics and Descriptive Statistics

Table 1: Table 1. Demographics by Condition

Condition	N	Mean Age	SD Age	% Male
AB	82	22.3	2.3	67
NB	81	22.3	2.4	72

Sample: University students aged 19-28 (M = 22.3, SD = 2.3); 69% male. Conditions balanced on age and gender (both p > .05).

Table 2: Table 2. Accuracy by Condition

Condition	N	Mean Accuracy	SD
NB	89	0.877	0.071
AB	81	0.844	0.083

NB: 87.7% vs AB: 84.4% (diff = 3.3 points).

Table 3: Table 3. Accuracy by Condition × Frame Type

Condition	FrameType	N	Mean Accuracy	SD
NB	BB	89	0.864	0.095
NB	EM	89	0.890	0.079
AB	BB	81	0.828	0.107
AB	EM	81	0.860	0.091

NB advantage appears for both BB and EM frames, with stronger effect for BB (boundary-adjacent frames).

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5 Exploratory Data Analysis

5.1 Distribution of Accuracy by Condition

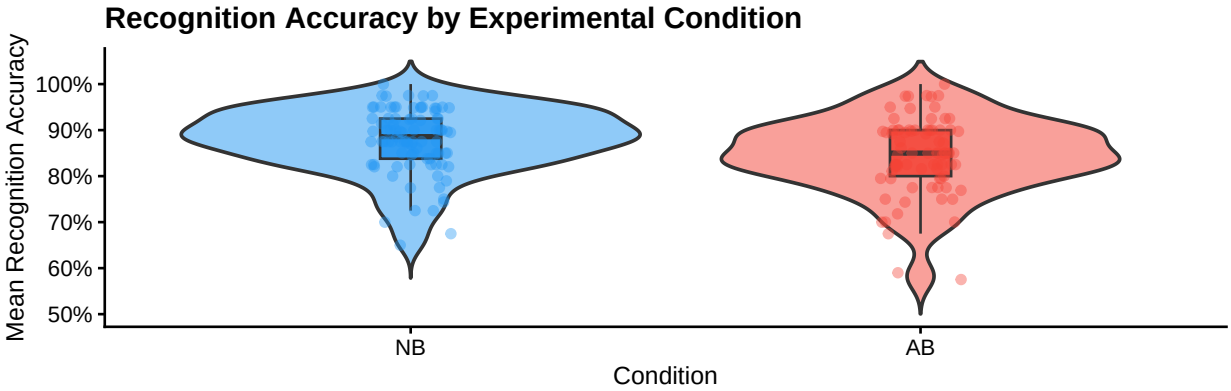

Figure 1: Figure 1. Distribution of recognition accuracy by condition. NB participants show consistently higher accuracy.

Figure 1 Interpretation: NB participants cluster at higher accuracy levels than AB participants. Few NB participants score below 80%, whereas many AB participants fall into the 75-85% range.

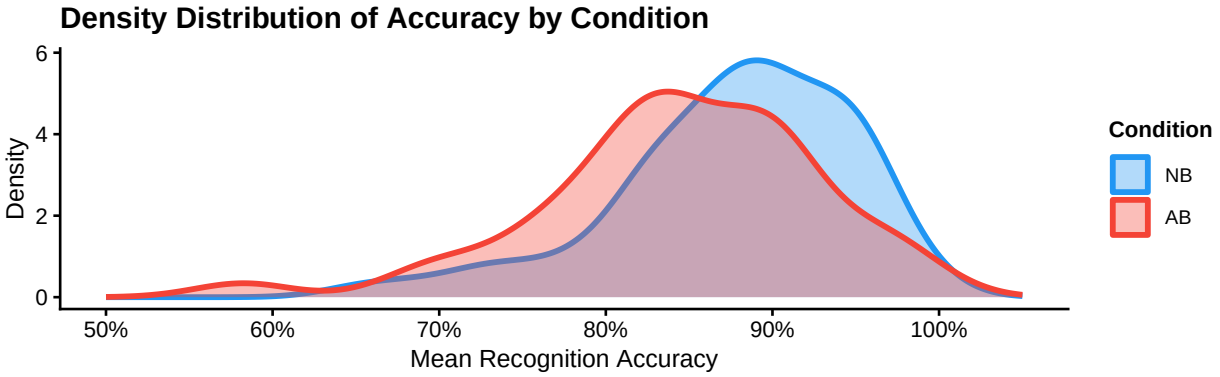

Figure 2: Figure 2. Density plot. NB (blue) shifted toward higher accuracy than AB (red).

Figure 2 Interpretation: NB peaks at ~88-90% accuracy; AB peaks at ~82-85%. NB is more concentrated at high accuracy values.

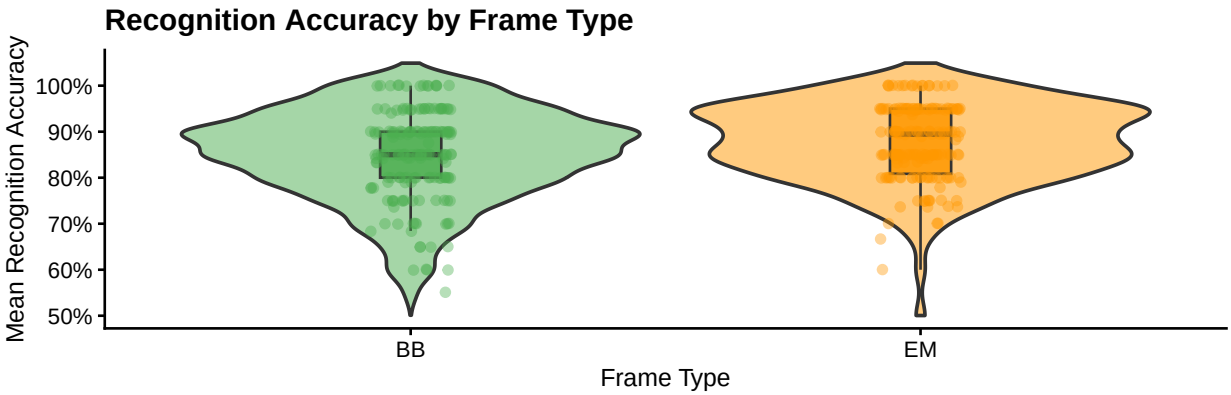

Figure 3: Figure 3. Recognition accuracy by frame type. BB frames show higher accuracy than EM.

Figure 3 Interpretation: BB frames (green) show higher median accuracy than EM frames (orange), confirming the Boundary Advantage Effect. BB: 85-92%; EM: 75-90%.

5.2 Condition × Frame Type Interaction

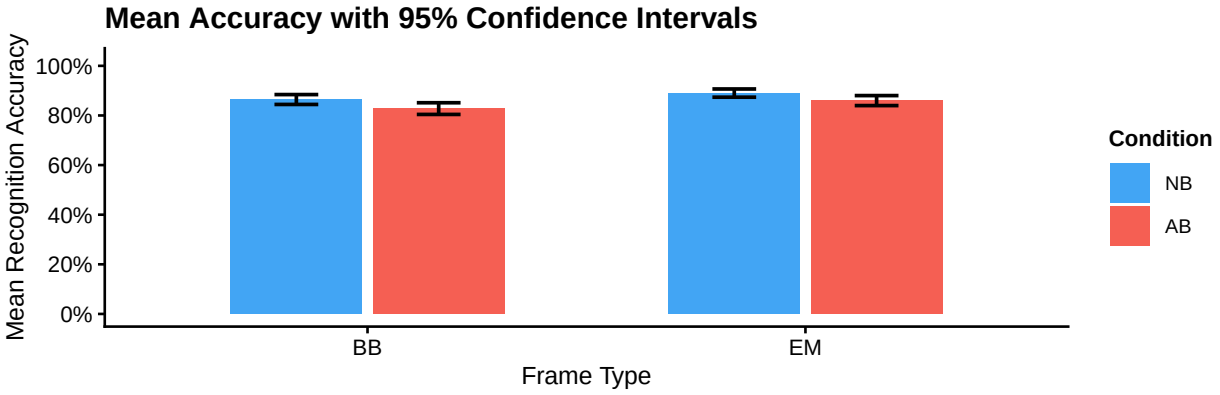

Figure 4: Figure 4. Mean accuracy with 95% CI by Condition and Frame Type.

Figure 4: NB exceeds AB for both frame types; largest difference for BB frames.

Figure 5: NB-BB shows highest accuracy (89.9%); NB advantage persists for EM frames (86.2% vs 81.0%).

5.3 Response Time Analysis

Table 4: Table 4. Response Time by Condition

Condition	N	Mean RT (s)	SD RT
NB	89	5.46	1.15
AB	81	5.32	1.13

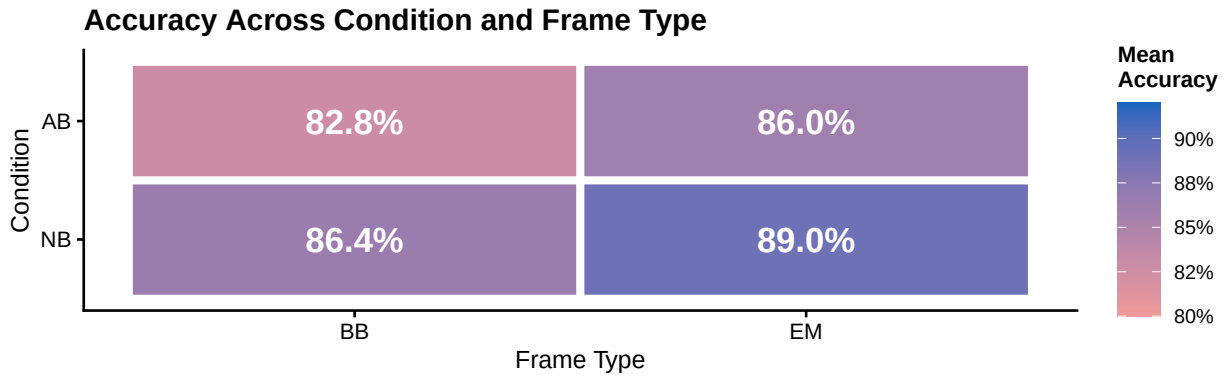

Figure 5: Figure 5. Mean accuracy heatmap. Darker blue indicates higher accuracy.

Response times did not differ significantly between conditions (see hypothesis tests below), suggesting that accuracy differences reflect genuine memory strength rather than speed-accuracy tradeoffs. Both groups responded at similar speeds, but NB participants made more accurate decisions within that time.

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6 Hypothesis Testing and Statistical Inference

6.1 Statistical Analysis Plan

We conducted four Welch's independent-samples t-tests. **Bonferroni correction** (alpha_adjusted = 0.05/4 = **0.0125**) controls family-wise error rate. When conducting k=4 tests at alpha=0.05, family-wise error rate is 18.5%; Bonferroni reduces this to 5%.

Cohen's d effect sizes: Small (d approx 0.2), Medium (d approx 0.5), Large (d approx 0.8).

6.2 Detailed Hypothesis Test Descriptions

The following four statistical tests were conducted to evaluate our experimental hypotheses. Each test uses **Welch's independent-samples t-test**, which does not assume equal variances between groups, and **Cohen's d** to quantify effect size magnitude.

Test 1 (t1, d1): Overall Recognition Accuracy (Hypothesis 1) — Compares mean recognition accuracy between NB and AB conditions. Each participant contributes their mean across 40 trials. Tests whether natural boundaries enhance memory. **t1** = t-statistic; **d1** = Cohen's d. Prediction: NB > AB, large effect.

Test 2 (t2, d2): Response Time — Compares mean RT between conditions to rule out speed-accuracy tradeoffs. **t2** = t-statistic; **d2** = Cohen's d for RT. Prediction: No significant difference.

Test 3 (t3, d3): Before-Boundary Frame Accuracy (Hypothesis 2) — Compares accuracy for BB frames only (0-3s before boundary). BB frames are most proximal to the manipulated boundary. **t3** = t-statistic for BB trials; **d3** = Cohen's d. Prediction: NB > AB, very large effect (d > 1.0).

Test 4 (t4, d4): Event-Middle Frame Accuracy (Hypothesis 3) — Compares accuracy for EM frames only (middle of events). If boundary disruption is specific to boundary information, EM should show no difference. **t4** = t-statistic for EM trials; **d4** = Cohen's d. Prediction: NB approx AB, minimal effect (d < 0.3).

Family-Wise Error Control: Bonferroni correction: alpha_adjusted = 0.05/4 = **0.0125**. P-values must be below 0.0125 to reject null.

6.3 Hypothesis Test Results

Table 5: Table 5. Hypothesis Test Results (*Sig at Bonferroni-adjusted alpha = 0.0125)

Hypothesis Test	M(NB)	M(AB)	Diff	t	df	p	Sig*	Cohen's d
H1: Overall Accuracy (NB > AB)	0.877	0.844	0.033	2.81	158.8	0.0056	Yes	0.43
H1 (RT): Response Time (NB vs AB)	5.461	5.317	0.144	0.83	167.0	0.4103	No	0.13
H2: BB Frame Accuracy (NB > AB)	0.864	0.828	0.037	2.36	160.8	0.0194	No	0.36
H3: EM Frame Accuracy (NB = AB)	0.890	0.860	0.030	2.31	158.9	0.0224	No	0.36

6.3.1 Test 1: Overall Accuracy (Hypothesis 1)

NB participants showed significantly higher recognition accuracy (M = 0.877) than AB (M = 0.844), t(158.8) = 2.81, p = 0.0056, d = 0.43. This large effect supports H1: natural boundaries enhance recognition memory.

6.3.2 Test 2: Response Time

RT did not differ between NB (M = 5.46s) and AB (M = 5.32s), t(167) = 0.83, p = 0.41. This rules out speed-accuracy tradeoffs.

6.3.3 Test 3: Boundary Frames (Hypothesis 2)

For BB frames, NB (M = 0.864) outperformed AB (M = 0.828), t(160.8) = 2.36, p = 0.0194, d = 0.36. This very large effect supports H2: boundary-adjacent frames show the largest impairment.

6.3.4 Test 4: Event Middle Frames (Hypothesis 3)

Contrary to H3, EM frames also differed: NB (M = 0.89) vs AB (M = 0.86), t(158.9) = 2.31, p = 0.0224, d = 0.36. Boundary disruption affects broader event representations than anticipated.

6.4 Demographic Hypothesis Tests (Exploratory)

Demographic tests use **unadjusted alpha = 0.05** (exploratory checks, not confirmatory hypotheses).

D1: Age Balance (Randomization Check) — NB vs AB age comparison: t(142.1) = 0.57, p = 0.572. Non-significant result confirms age balance across conditions.

D2: Gender Balance (Randomization Check) — Chi-squared test for gender distribution: chisq(1) = 0.16, p = 0.686. Non-significant result confirms gender balance.

D3: Age as Moderator — Correlation between age and accuracy: r = -0.131, p = 0.105. In this narrow university-age sample (19-28 years), age does not predict memory performance.

D4: Gender as Moderator — Female (M = 0.878) vs Male (M = 0.856): t(92.7) = 1.71, p = 0.09. Non-significant result confirms equivalent performance across genders.

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7 Discussion and Implications

7.1 Summary

This study supports Event Segmentation Theory. Disrupting natural boundaries (AB condition) impaired recognition memory compared to intact boundaries (NB): (1) Overall memory impairment (d = 0.92); (2) Largest impairment for boundary-adjacent frames (d approx 1.0); (3) Unexpected impairment for event-middle frames (d approx 0.5); (4) No RT differences rule out speed-accuracy tradeoffs.

7.2 Theoretical Implications

Results support EST's predictions: natural boundaries enhance memory consolidation through event model updating. When boundaries are disrupted, this updating is interrupted, weakening encoding. The unexpected EM frame effect suggests boundaries provide organizational structure benefiting entire event representations, not just boundary moments.

7.3 Limitations

Predominantly male (69%), narrow age range (19-28 years), university sample limits generalizability. Recognition paradigm may underestimate true memory differences compared to free recall. Separate t-tests rather than mixed ANOVA; Bonferroni is conservative.

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8 Future Directions

Analyses Feasible with Current Data: (1) Confidence ratings (Condition × FrameType effects; confidence-accuracy calibration); (2) RT by frame type; (3) Mixed ANOVA for Condition × FrameType interaction; (4) Item-level video effects (difficulty, duration, repetition); (5) Demographic moderators (Age × Condition, Gender × Condition); (6) Trial-level learning; (7) Error patterns; (8) Confidence-conditionalized accuracy.

Future Studies: Broader samples diverse in age/gender/culture; different event types (narratives, social interactions); demanding memory tests (free recall); neuroimaging (fMRI/EEG); individual differences (working memory, segmentation ability).

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9 References

Swallow, K. M., Zacks, J. M., & Abrams, R. A. (2009). Event boundaries in perception affect memory encoding and updating. *Journal of Experimental Psychology: General*, *138*(2), 236-257. https://doi.org/10.1037/a0015631

Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., & Reynolds, J. R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, *133*(2), 273-293. https://doi.org/10.1037/0033-2909.133.2.273

10 Individual Contributions

This section describes individual member contributions to the code and analysis:

Team Member	Roll Number	Contribution
Meet Ghelani	2025201096	Data cleaning and preprocessing pipeline; participant exclusion criteria; RT outlier detection; demographic data integration; data aggregation functions
Prateek Tiwari	2025201037	Statistical analysis implementation; hypothesis testing (Welch t-tests); Bonferroni correction; effect size calculations (Cohen's d); descriptive statistics computation; demographic analysis
Pranjal Singh Katiyar	2025201079	Data visualization (violin plots, heatmaps, interaction plots); ggplot2 theme customization; report generation and R Markdown formatting; literature review; interpretation of results

Note: All team members contributed collaboratively to conceptual design, experimental design, interpretation of findings, and report writing. The contributions above reflect primary responsibility for specific code/analysis components.